

JUNYING WANG

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EDUCATION

University of Southern California, Los Angeles, CA PhD Candidate in Computer Science, advised by <u>Prof. Ulrich Neumann</u> and <u>Prof. Yue Wang</u>	08/2020 - Present
University of Southern California, Los Angeles, CA Master of Engineering in Computer Science, Advisor: <u>Prof. Hao Li</u>	01/2018 - 05/2020
Ocean University of China, Qingdao, China Bachelor of Engineering in Computer Science	06/2013 - 07/2017

RESEARCH AREAS & SKILLS

Research Areas:	Digital Human, Generative AI, 3D Recontruction, Neural Rendering, Humanoid
Programming:	Python, C/C++, Java
Software & Tools:	PyTorch, OpenGL, Blender, Maya, Unity

WORK EXPERIENCE

Meta Inc. Research Scientist Intern - Mentor: <u>Dr. Yuanlu Xu</u> • Work on the research project - Generalizable Human 3D Gaussian Splatting Editing and Enhancement. • Exploring multi-layer Gaussian avatar editing and enhancement using 3D generative models. • Contribute one submission to CVPR 2026 (under review).	05/2025 - 12/2025
Meta Inc. Research Scientist Intern - Mentor: Dr. Tony Tung • Work on the research project - Dynamic Gaussian Hair. • Develop a data-driven dynamic hair solver for modeling and rendering dynamic hair using 3D Gaussian representations. • Contribute one submission to NeurIPS 2025 (published).	06/2024 - 12/2024
Adobe Inc. Research Scientist Intern - Mentor: <u>Dr. Jae Shin Yoon</u> • Work on the research project - Repurposing Diffusion Models for Generalizable and Consistent Monocular Human Relighting. • Propose a novel method that can control the light from an image or video of humans with arbitrary body parts. • Contribute one submission to CVPR 2025 (published).	09/2023 - 01/2024
Adobe Inc. Research Scientist Intern - Mentor: <u>Dr. Jae Shin Yoon</u> • Work on the research project - Complete 3D Human Reconstruction from a Single Incomplete Image. • Build a novel model to reconstruct a complete human geometry and texture from a partial body image. • Contribute one submission to CVPR 2023 (published).	05/2022 - 12/2022

PUBLISHED PUBLICATIONS

• DGH: Dynamic Gaussian Hair <i>Junying Wang, Yuanlu Xu, Edith Tretschk, Ziyang Wang, Anastasia Ianina, Aljaž Božič, Ulrich Neumann, Tony Tung</i> <i>Conference on Neural Information Processing Systems (NeurIPS), 2025</i>	2025
• Comprehensive Relighting: Generalizable and Consistent Monocular Human Relighting and Harmonization <i>Junying Wang, Jingyuan Liu, Xin Sun, Krishna Kumar Singh, Zhixin Shu, He Zhang, Jimei Yang, Nanxuan Zhao, Tuanfeng Yang Wang, Simon Su Chen, Ulrich Neumann, Jae Shin Yoon</i> <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025</i>	2025
• Boosting Generalizability towards Zero-Shot Cross-Dataset Single-Image Indoor Depth by Meta-Initialization <i>Cho-Ying Wu, Yiqi Zhong, Junying Wang, Ulrich Neumann</i> <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024</i>	2024
• Complete 3D Human Reconstruction from a Single Incomplete Image <i>Junying Wang, Jae Shin Yoon, Tuanfeng Wang, Krishna Kumar Singh, Ulrich Neumann</i> <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023</i>	2023
• Meta-Optimization for Higher Model Generalizability in Single-Image Depth Prediction <i>Cho-Ying Wu, Yiqi Zhong, Junying Wang, Ulrich Neumann</i> <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023</i>	2023

PATENT

- **Complete 3D Object Reconstruction from an Incomplete Image** **filed 03/06/2025**
Jae Shin Yoon, Tuanfeng Wang, Krishna Kumar Singh, Junying Wang, Jingwan Lu
U.S. Patent Application No. 18/242,380
- **Systems and Methods for Generating a Relighted Image** **filed 01/22/2026**
Junying Wang, Jae Shin Yoon, Jingyuan Liu, Xin Sun, Krishna Kumar Singh, Zhixin Shu, He Zhang, Jimei Yang, Yangtuanfeng Wang, Nanxuan Zhao, Simon Chen
U.S. Patent Application No. 18/775,063

SELECTED PROJECTS

- Humanoid Everywhere: A Scalable Environment and Data Generator for Humanoid Manipulation** **12/2025 – Present**
 - A research project on scalable generation of simulation-ready 3D environments and assets for humanoid manipulation training.
 - Aiming to generate large-scale digital assets from text prompts as a generative prior for real-world deployment.
 - Phase I: Text-to-3D asset generation producing physically stable scenes and objects for post-training.
 - Phase II: Integration of generated assets into a simulator for training humanoid grasping and motion planning policies.
- Free-View Volumetric Human Body Capturing From Sparse Views** **05/2021 - 11/2021**
 - A research project presents a novel end-to-end free-view human performance capturing approach from sparse views.
 - The first non-parametric human body capturing system using volumetric rendering with only 2D image supervision.
 - Built a novel end-to-end trainable sparse-view volumetric human body capturing system for high-quality, free-view rendering.
 - Proposed a novel contrastive learning approach to ensure the local smoothness and continuity of human body surface.
- Contrastive Learning for Image Disentanglement** **01/2020 -06/2020**
 - A research project of training a general model for image disentanglement by using contrastive learning.
 - Via user studies, the proposed model can do feasible image disentanglement on different datasets.
 - Proposed an intuitive approach of unsupervised end-to-end learning for image disentanglement.
 - Proved that using contrastive learning on image features can enhance the accuracy and fidelity of the generated images.
- Img2Motion: Learning to Drive 3D Avatar Using Videos** **05/2019 -11/2019**
 - A research project of building a neural network motion retargeting system.
 - Via user studies, the designed system can successfully animate any 3D avatar using the motions extracted from 2D videos.
 - Proposed a neural network motion retargeting system that utilizes videos to drive 3D subjects with varying skeleton structures.
 - Built a large-scale dataset with varieties of human poses by animating existing rigged human models.
- K-D Tree for Faster Path Tracer** **10/2019 - 11/2019**
 - A personal project of using k-d tree to do faster path tracing.
 - Compared to traditional path tracer, this render can cut in half of the running time.
 - Completed reproducing a more complex light environment through the use of Monte Carlo integration.
 - Used bounding boxes to represent objects; used k-d tree to store bounding boxes to achieve running time improvements.

PROFESSIONAL SERVICES

- Teaching Experience**
 - Teaching Assistant, USC CSCI-570 Analysis of Algorithms (Instructor: Prof. Victor Adamchik) **01/2021 – 05/2021**
 - Teaching Assistant, USC CSCI-576 Multimedia Systems Design (Instructor: Prof. Parag Havaladar) **01/2022 – 12/2025**
- Conference Reviewer**
 - Computer Vision and Pattern Recognition (CVPR) **2024 - 2026**
 - International Conference on Robotics and Automation (ICRA) **2026**
 - European Conference on Computer Vision (ECCV) **2026**
 - International Conference on Computer Vision (ICCV) **2025**
 - ACM International Conference on Multimedia (MM) **2025**
 - The AAAI Conference on Artificial Intelligence (AAAI) **2025-2026**
 - International Conference on 3D Vision (3DV) **2025-2026**
- Mentorship & Outreach**
 - USC Graduate Student Mentorship Program (Mentor) **2025-2026**
 - USC Women in Engineering Mentorship Program (Mentor) **2025-2026**
- Honors & Awards**
 - Outstanding Undergraduate Scholarship, Ocean University of China (OUC) **2014-2016**
 - USC Graduate Student Government (GSG) Professional Development Fund **2023, 2025**